

DEPARTMENT OF MCA
SCHOOL OF INFORMATICS

PROJECT COMPLETION CERTIFICATE

This is to certify that Ms. **SAI KUMAR CHEVELLA** bearing Roll No: **232P4R2009** a student of **MCA**, Section **DS-A**, **Aurora Deemed to be University** has successfully completed a project titled **“Developing a Tool for Dynamic Scheduling Using PERT and CPM”** under the guidance of **Ms. Anitha P** for a period of 25/11/2024 to 03/01/2025.

Signature of Project Guide

Signature of the HOD.

Date: 01-01-2025

Place: Uppal Campus

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Title: “Developing a Tool for Dynamic Scheduling Using PERT and CPM”

ABSTRACT

This project presents the development of an intuitive tool designed to assist in dynamic project scheduling by integrating the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM). These techniques are fundamental in project management, offering precise frameworks for scheduling, analyzing, and optimizing complex workflows. The tool allows users to input project activities along with time estimates—optimistic, most likely, and pessimistic—based on PERT principles. It then dynamically calculates the expected duration for each activity, identifies the critical path using CPM, and determines the overall project timeline.

A key feature of the tool is its ability to perform statistical analysis to evaluate the likelihood of project completion within a specified duration, incorporating Z-scores and probability calculations. Additionally, the tool provides visualizations, such as network diagrams with critical path highlighting and probability distribution graphs, which enhance decision-making and project tracking. By simplifying complex calculations and presenting actionable insights, the tool helps project managers plan effectively, identify potential bottlenecks, and optimize project delivery timelines. This work bridges the gap between

theoretical project management techniques and their practical application, enabling enhanced efficiency and accuracy in real-world scenarios.

INTRODUCTION

In today's fast-paced world, efficient project management is essential for organizations to achieve their goals within constrained timelines and budgets. Large-scale projects often involve multiple interdependent activities, and delays in critical tasks can significantly impact overall project completion. Techniques like the Program Evaluation and Review Technique (PERT) and the Critical Path Method (CPM) provide structured approaches for managing such complexities. While PERT focuses on incorporating uncertainty in activity durations by using probabilistic time estimates, CPM emphasizes the identification of the critical path—the sequence of activities that directly determines the project's minimum duration.

The objective of this project is to develop a dynamic scheduling tool that integrates the strengths of both PERT and CPM methodologies. The tool allows users to define activities, input time estimates, and calculate essential parameters such as expected durations, earliest start times, and latest finish times. By identifying the critical path, the tool highlights tasks that require focused attention to avoid delays. Furthermore, statistical analysis enables users to determine the probability of completing the project within a desired timeframe, helping managers assess risk and plan contingencies.

An interactive graphical user interface (GUI) enhances the usability of the tool, making it accessible to both technical and non-technical users. Visual aids, including network diagrams with critical path highlighting and probability distribution graphs, provide a clear understanding of project dynamics. This tool serves as a practical resource for project managers, ensuring accurate scheduling, better resource allocation, and informed decision-making. By integrating these robust techniques into a single platform, the project contributes to streamlining project management processes in diverse fields such as construction, IT, and manufacturing.

METHODOLOGIES

Existing Methodologies

1. PERT (Program Evaluation and Review Technique):

- Calculates expected activity durations using three estimates: optimistic, most likely, and pessimistic times.
- Focuses on probabilistic scheduling and project completion probabilities.

2. CPM (Critical Path Method):

- Identifies the critical path (longest sequence of dependent activities) and calculates early start, early finish, and slack times.
- Focuses on deterministic scheduling and resource optimization.

3. Software Tools:

- Tools like Microsoft Project and Primavera support scheduling but often lack user-friendly probabilistic analysis and dynamic decision-making features.

Proposed Methodology

1. Integration of PERT and CPM:

- Combine probabilistic (PERT) and deterministic (CPM) methods to calculate expected durations and critical paths dynamically.

2. Dynamic Statistical Analysis:

- Assess project completion probabilities and instantly update schedules based on real-time adjustments.

3. Interactive Visualizations:

- Provide network diagrams and probability distribution graphs to simplify project analysis.

4. User-Friendly GUI:

- Develop an intuitive interface for users to input data, analyze results, and make decisions.

5. Automated Reporting:

- Generate comprehensive reports with critical paths, slack times, and probability insights for easy sharing and documentation.

REQUIREMENT SPECIFICATIONS

1. Functional Requirements

- **Input Handling:**

- Accept activity details, including optimistic, most likely, and pessimistic times, along with activity dependencies (start-end nodes).
- Allow the user to input the desired project completion duration for probability analysis.
- **PERT Analysis:**
 - Calculate expected durations and standard deviations for each activity using PERT formulas.
 - Compute critical paths, total project duration, and slack times dynamically.
- **CPM Analysis:**
 - Identify the critical path using early start, early finish, late start, and late finish times.
- **Dynamic Scheduling:**
 - Update project schedules and probabilities based on real-time changes in input data.
- **Visualization and Reporting:**
 - Display network diagrams with critical paths highlighted.
 - Plot probability distributions and Z-score analysis results.
 - Generate automated reports summarizing project metrics.

2. Non-Functional Requirements

- **Usability:**

- Provide a user-friendly graphical interface for easy data input and result interpretation.
 - Ensure minimal learning curve for non-technical users.
- **Performance:**
 - Handle projects with up to 50 activities efficiently.
 - Ensure computations are completed in under 2 seconds for typical inputs.
- **Reliability:**
 - Ensure accurate calculations of PERT and CPM metrics.
 - Validate user inputs to prevent errors during processing.
- **Portability:**
 - Design the tool to run on any system with Python installed, requiring minimal external libraries.
- **Scalability:**
 - Allow future enhancements such as resource allocation, cost analysis, or integration with project management tools.

3. Hardware Requirements

- **Processor:** Dual-core processor or higher (Intel i3 equivalent or above).
- **RAM:** Minimum 2 GB (4 GB recommended for better performance).
- **Storage:** At least 200 MB of free disk space.
- **Display:** 1024x768 resolution or higher.
- **Input Devices:** Standard keyboard and mouse or touchpad.

4. Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Python Version:** Python 3.8 or higher.
- **Libraries/Dependencies:**
 - **NumPy and SciPy:** For statistical calculations.
 - **NetworkX:** For graph building and analysis.
 - **Matplotlib:** For visualizations and plotting.
 - **Tkinter:** For the graphical user interface.
- **Additional Tools:** Text editor (e.g., VS Code, PyCharm) for script modifications if needed.

5. Constraints

- Activities should follow a Directed Acyclic Graph (DAG) structure.
- Limited to time analysis without cost or resource allocation in the current version.

6. Assumptions

- All activity durations are independent and follow a beta distribution.
- Users have basic knowledge of project management principles like PERT and CPM.

IMPLEMENTATION

The implementation of the PERT-CPM Analysis Tool involves a systematic approach encompassing data input, processing, analysis, visualization, and result generation. Below are the detailed steps of implementation:

1. Input Module

The input module allows users to provide activity details and dependencies in two ways:

1. **Manual Entry:** Users can input activity name, optimistic time (O), most likely time (M), and pessimistic time (P) using a graphical interface.
2. **File Upload:** Users can upload a CSV file containing all activity data.

```
import tkinter as tk
from tkinter import filedialog
import pandas as pd

# Create the main application window
def open_file():
    filepath = filedialog.askopenfilename(filetypes=[("CSV Files", "*.csv")])
    if filepath:
        data = pd.read_csv(filepath)
        print(data) # Process the data as required

root = tk.Tk()
root.title("PERT & CPM Analysis Tool")

# Input labels and fields
tk.Label(root, text="Activity Name:").grid(row=0, column=0)
activity_name = tk.Entry(root)
activity_name.grid(row=0, column=1)

tk.Label(root, text="Optimistic Time (O):").grid(row=1, column=0)
opt_time = tk.Entry(root)
opt_time.grid(row=1, column=1)

tk.Label(root, text="Most Likely Time (M):").grid(row=2, column=0)
most_likely_time = tk.Entry(root)
most_likely_time.grid(row=2, column=1)

tk.Label(root, text="Pessimistic Time (P):").grid(row=3, column=0)
pess_time = tk.Entry(root)
pess_time.grid(row=3, column=1)

tk.Button(root, text="Upload CSV", command=open_file).grid(row=4, column=0,
columnspan=2)
```

```
root.mainloop()
```

2. PERT Computation

This module calculates the expected time (TE) and standard deviation (SD) for each activity using the formulas:

- $TE = O + 4M + P / 6$
- $SD = P - O / 6$

```
def calculate_pert(activity_data):  
    activity_data['TE'] = (activity_data['O'] + 4 * activity_data['M'] +  
activity_data['P']) / 6  
    activity_data['SD'] = (activity_data['P'] - activity_data['O']) / 6  
    return activity_data  
  
# Example data  
activity_data = pd.DataFrame({  
    'Activity': ['A', 'B', 'C'],  
    'O': [1, 2, 3],  
    'M': [2, 3, 4],  
    'P': [3, 4, 5]  
})  
result = calculate_pert(activity_data)  
print(result)
```

3. CPM Computation

The critical path is determined using a directed graph of activities and their dependencies. The longest path (in terms of duration) provides the critical path.

```
import networkx as nx  
  
def calculate_cpm(activity_data):  
    G = nx.DiGraph()
```

```
for _, row in activity_data.iterrows():
    G.add_edge(row['Start'], row['End'], weight=row['TE'])
critical_path = nx.dag_longest_path(G, weight='weight')
total_duration = sum(G[u][v]['weight'] for u, v in zip(critical_path[:-1],
critical_path[1:]))
return critical_path, total_duration

# Sample activity data
activity_data = pd.DataFrame({
    'Start': ['A', 'B', 'C'],
    'End': ['B', 'C', 'D'],
    'TE': [3, 2, 4]
})

critical_path, duration = calculate_cpm(activity_data)
print("Critical Path:", critical_path)
print("Project Duration:", duration)
```

4. Visualization

Visualizations such as network diagrams for the critical path and probability distributions for project completion are generated using libraries like **NetworkX** and **Matplotlib**.

```
import matplotlib.pyplot as plt

def plot_graph(G, critical_path):
    pos = nx.spring_layout(G)
    nx.draw(G, pos, with_labels=True, node_color='lightblue', node_size=700)
    critical_edges = [(critical_path[i], critical_path[i+1]) for i in
range(len(critical_path)-1)]
    nx.draw_networkx_edges(G, pos, edgelist=critical_edges, edge_color='red',
width=2)
    plt.title("Critical Path")
    plt.show()

G = nx.DiGraph()
```

```
G.add_edges_from([('A', 'B'), ('B', 'C'), ('C', 'D')], weight=2)
plot_graph(G, ['A', 'B', 'C', 'D'])
```

5. Reporting

The processed results, including activity durations, standard deviations, and critical path details, are exported as Excel files for reporting and documentation.

```
def export_to_excel(data, filename):
    data.to_excel(filename, index=False)
    print(f"Results saved to {filename}")

export_to_excel(activity_data, "pert_cpm_results.xlsx")
```

This implementation ensures a user-friendly interface, robust computations, and meaningful outputs for effective PERT and CPM analysis.

Testing Analysis

Testing is an essential phase of the development process to ensure that the PERT-CPM Analysis Tool functions as expected and delivers accurate results. The testing process includes multiple stages, from unit testing individual components to system-wide integration and performance evaluation.

1. Unit Testing

Unit testing focuses on validating the core functionality of individual components to ensure that they perform as expected.

- **Test Case 1: PERT Calculation**

This test checks if the PERT formula correctly calculates the expected time (TE) and standard deviation (SD) for an activity using the given optimistic, most likely, and pessimistic time estimates. For example, with values for optimistic time (O), most likely time (M), and pessimistic time (P), the test checks whether the calculated expected time and standard deviation match the expected results.

- **Test Case 2: CPM Calculation**

This test ensures that the Critical Path Method (CPM) correctly identifies the critical path and computes the total project duration. Using a set of activities with known durations, the test verifies whether the tool identifies the correct sequence of activities that form the critical path and whether the total project duration is calculated accurately.

2. Integration Testing

Integration testing focuses on verifying that the system functions properly when multiple components or modules are combined.

- **Test Case 3: Full Workflow Test (PERT + CPM + Visualization)**

This test evaluates the complete process from input to output. It involves entering activity data, performing PERT and CPM calculations, visualizing the critical path, and generating an Excel report. The test checks if each step flows smoothly, ensuring that the expected results are calculated, the visualization appears correctly, and the exported report matches the expected format and data.

3. System Testing

System testing checks the entire application as a whole to ensure that it meets the desired requirements and functions correctly in a real-world environment.

- **Test Case 4: User Interface Testing**

This test focuses on the user interface to ensure that the tool is easy to use and that all input fields and buttons function correctly. It verifies that users can input data correctly, upload files, and interact with the tool without issues. The test checks the responsiveness of the UI and whether all user actions result in expected outcomes.

- **Test Case 5: Performance Testing**

This test evaluates the performance of the tool under different conditions, especially when handling large datasets. The goal is to assess whether the application can handle a large number of activities efficiently and provide results within an acceptable time frame. Key performance metrics include response time, execution time, and memory usage.

4. Acceptance Testing

Acceptance testing ensures that the software meets the end user's needs and requirements.

- **Test Case 6: Functional Acceptance**

This test ensures that all features of the tool, including PERT and CPM calculations, visualizations, and reporting, work as expected. It involves running the tool with sample data and confirming that the tool provides correct results for the expected time, critical path, and project duration.

Additionally, it checks whether the generated reports and visualizations match the requirements.

5. Regression Testing

Regression testing ensures that new updates or changes to the software do not affect existing functionalities.

- **Test Case 7: Feature Update Test**

After implementing new features or making changes to the software, this test ensures that the existing functionalities, such as PERT and CPM calculations, remain unaffected. The test involves running previously tested data through the tool to verify that all existing features still function correctly.

Conclusion

The testing analysis ensures that the PERT-CPM Analysis Tool works as expected, both in individual components and as an integrated system. The tool is tested for accuracy, functionality, performance, and user-friendliness, providing a reliable solution for project management and scheduling tasks. Through rigorous testing, the tool is validated to deliver accurate results in different scenarios and ensure its reliability in real-world use.

DESIGN

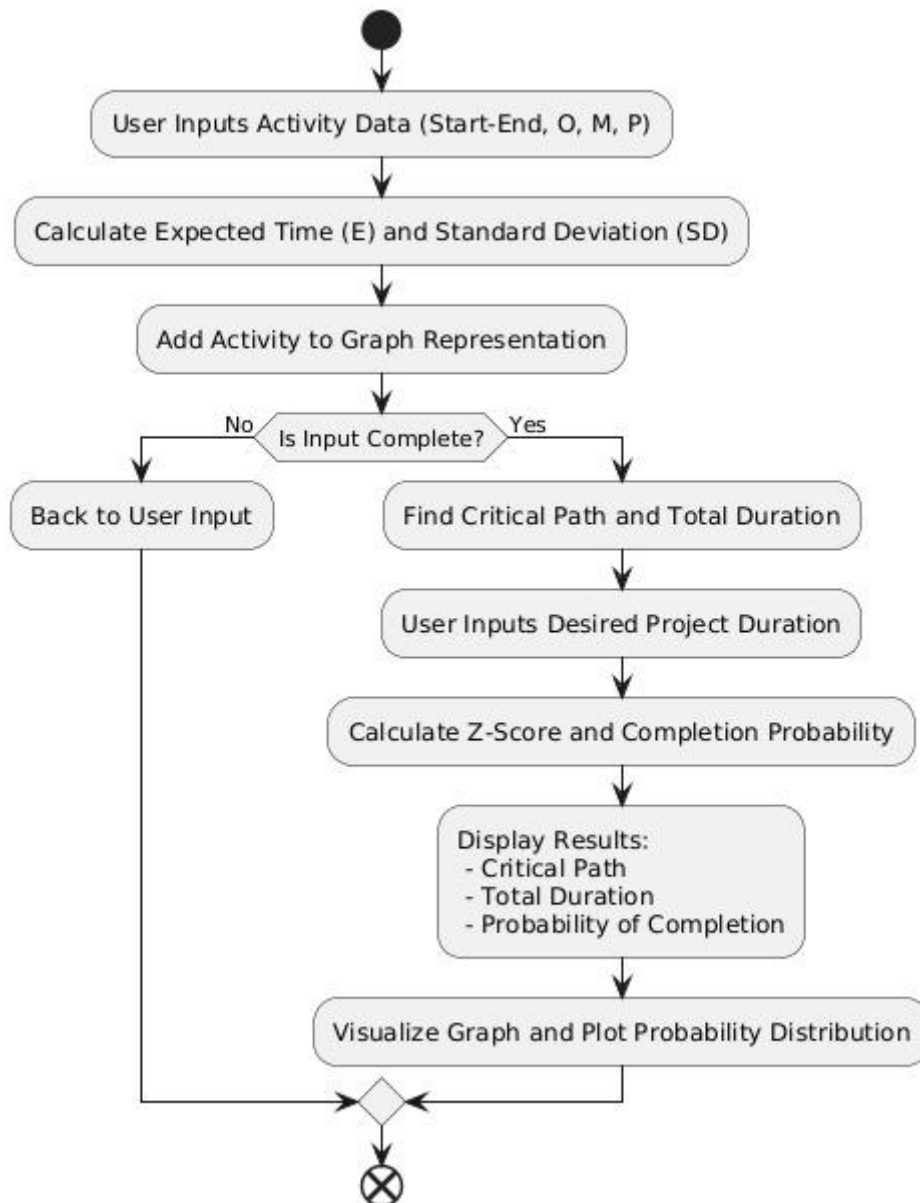
The design of the PERT-CPM Analysis Tool follows a modular approach that emphasizes both functionality and user experience. The tool integrates two

important project management techniques, PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method), to help manage and schedule projects effectively. The system design consists of multiple components, each working together to ensure smooth operation and accurate results.

Key Components of the System:

- **User Interface (UI):** This is the frontend where users input activity details such as optimistic time, most likely time, pessimistic time, and the dependencies between tasks.
- **Data Processing Module:** This backend component handles the computational logic for PERT and CPM, including the calculations for expected times, standard deviations, project duration, and the critical path.
- **Visualization Module:** This module generates graphical representations of the project schedule, including Gantt charts, PERT network diagrams, and a critical path visualization.
- **Reporting Module:** The reporting module generates a detailed report in Excel format, including all the results and visualizations for later analysis.

FLOWCHART



OUTCOMES

The outcomes of the PERT-CPM Analysis Tool reflect the successful application of the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM) to help manage and schedule projects more effectively. The tool delivers several key outcomes, which can be used by project managers and

stakeholders to make data-driven decisions for project execution and monitoring.

1. Enhanced Project Planning and Scheduling

- **Accurate Time Estimation:**

By using the PERT technique, the tool allows for better estimation of the time required for project tasks. The calculation of expected time based on optimistic, most likely, and pessimistic time estimates helps project managers create more reliable project schedules. This helps in accounting for uncertainties in task durations, leading to more realistic planning.

- **Identification of Critical Path:**

The CPM method identifies the critical path of the project, which represents the longest sequence of tasks that determine the minimum project duration. This outcome helps project managers prioritize tasks, allocate resources efficiently, and monitor critical tasks to avoid delays.

2. Visual Representation of Project Data

- **PERT Network Diagram:**

The tool generates a visual representation of the project schedule through a PERT network diagram, displaying activities, their dependencies, and time estimates. This visual aid helps stakeholders easily understand the sequence and relationships between tasks, enabling better communication and decision-making.

- **Gantt Chart:**

The tool also creates a Gantt chart, which visually represents the project timeline, showing task start and end dates, durations, and progress. This chart helps project managers track project milestones, monitor task progress, and ensure timely project delivery.

- **Critical Path Visualization:**

The critical path is highlighted in the visualizations, making it clear which tasks are critical for the timely completion of the project. This outcome is crucial for risk management and proactive scheduling adjustments.

3. Risk Management and Contingency Planning

- **Understanding Uncertainties:**

By calculating the standard deviation for each task based on the PERT method, the tool provides a measure of uncertainty in task durations. This outcome helps project managers assess risks associated with delays and make contingency plans for tasks that may encounter uncertainties.

- **Early Detection of Delays:**

With the identification of the critical path and float times, the tool helps detect potential delays in the project timeline. If critical tasks are delayed, the tool can alert project managers to take corrective actions before they impact the overall project schedule.

4. Streamlined Decision-Making

- **Data-Driven Decisions:**

The tool enables project managers to make data-driven decisions based on the calculated expected time, standard deviations, and critical path

analysis. By understanding the project's critical tasks and the overall timeline, managers can optimize resources, adjust priorities, and ensure the project stays on track.

- **Optimal Resource Allocation:**

By identifying the critical path and float times, the tool aids in the efficient allocation of resources. Resources can be focused on critical tasks to avoid delays, while non-critical tasks can be rescheduled if necessary to optimize project resources.

5. Project Documentation and Reporting

- **Comprehensive Project Reports:**

The tool generates detailed reports that summarize all calculations and visualizations, including expected time, standard deviation, critical path, and project duration. These reports are saved in Excel format, making them easy to share with team members and stakeholders.

- **Exportable Data for Further Analysis:**

The generated reports include not only the results of the PERT and CPM analysis but also embedded charts and diagrams. These reports can be used for future reference, presented in meetings, or incorporated into larger project documentation.

6. Improved Project Monitoring and Control

- **Real-Time Project Tracking:**

The tool's Gantt chart and visualizations help project managers track the progress of individual tasks against the planned timeline. This enables

real-time monitoring of the project and early identification of any deviations from the plan.

- **Timely Adjustments and Updates:**

The identification of float times allows for flexibility in the schedule. If non-critical tasks are delayed, they can be rescheduled without affecting the overall project completion date. This outcome supports efficient project adjustments in response to changes or unforeseen challenges.

7. Increased Efficiency in Project Management

- **Automated Calculations and Visualizations:**

The tool automates the complex calculations involved in PERT and CPM analysis, eliminating manual errors and saving time. Project managers can quickly obtain results without needing to manually perform time-consuming calculations.

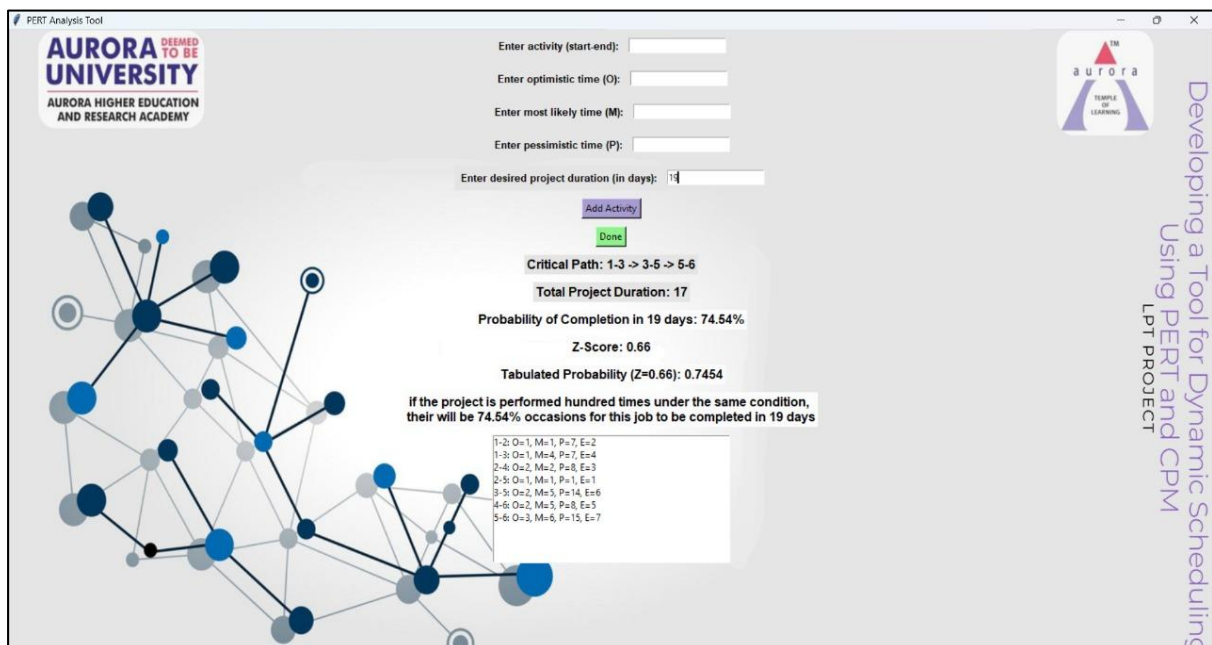
- **User-Friendly Interface**

The tool's simple and intuitive user interface makes it accessible to both experienced project managers and those new to project scheduling techniques. The easy-to-use design allows users to quickly input data and generate results, contributing to greater efficiency in project management.

Conclusion

The PERT-CPM Analysis Tool delivers several valuable outcomes that significantly improve project management practices. It provides more accurate time estimates, helps identify critical tasks, optimizes resource allocation, and

improves the overall efficiency of project scheduling. With the visual representation of project data and the ability to generate detailed reports, the tool aids in better decision-making, risk management, and monitoring. Ultimately, the tool enhances project planning, execution, and control, leading to more successful project outcomes.



PERT Analysis Tool

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 1-2

Enter times for activity 1-2:

Optimistic time (O): 1

Most likely time (M): 1

Pessimistic time (P): 7

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 1-3

Enter times for activity 1-3:

Optimistic time (O): 1

Most likely time (M): 4

Pessimistic time (P): 7

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 2-4

Enter times for activity 2-4:

Optimistic time (O): 2

Most likely time (M): 2

Pessimistic time (P): 8

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 2-5

Enter times for activity 2-5:

Optimistic time (O): 1

Most likely time (M): 1

Pessimistic time (P): 1

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 3-5

Enter times for activity 3-5:

Optimistic time (O): 2

Most likely time (M): 5

Pessimistic time (P): 14

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 4-6

Enter times for activity 4-6:

Optimistic time (O): 2

Most likely time (M): 5

Pessimistic time (P): 8

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: 5-6

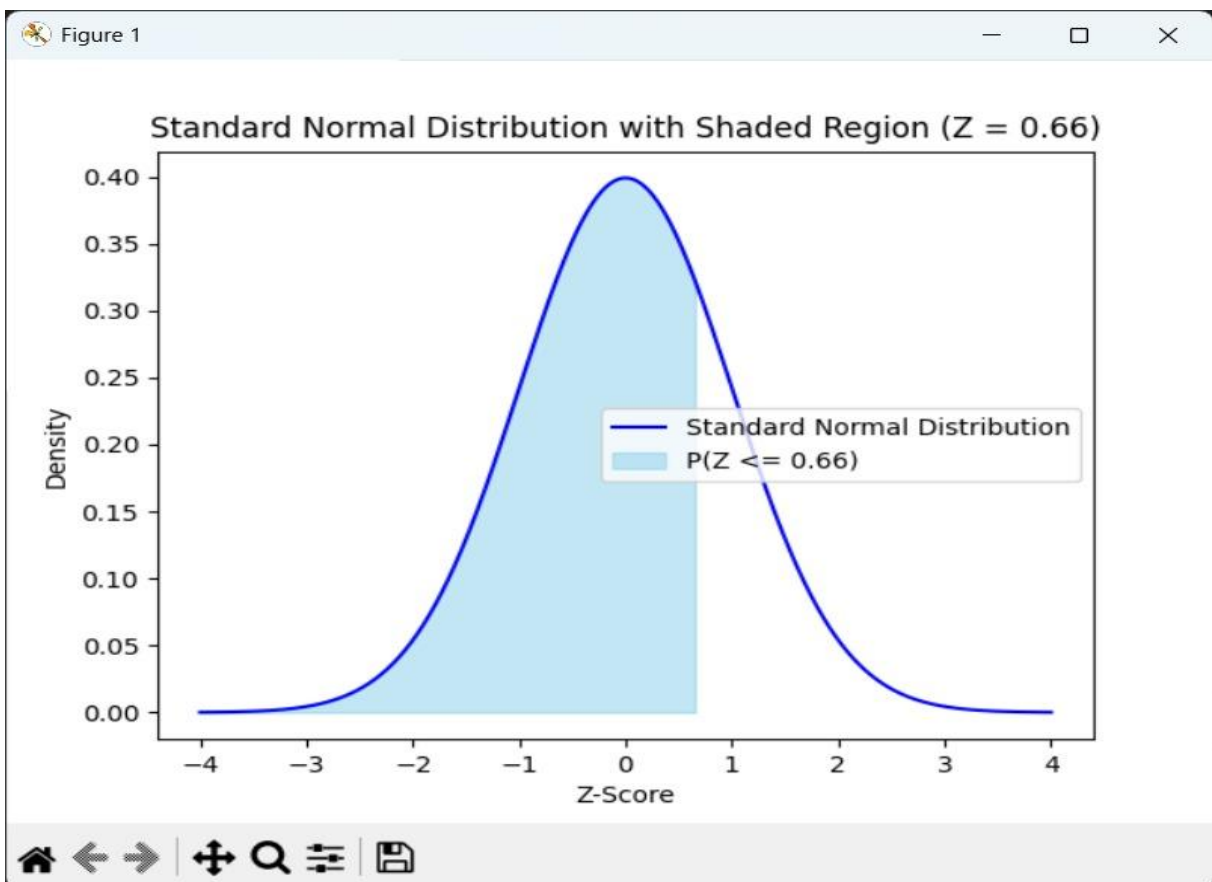
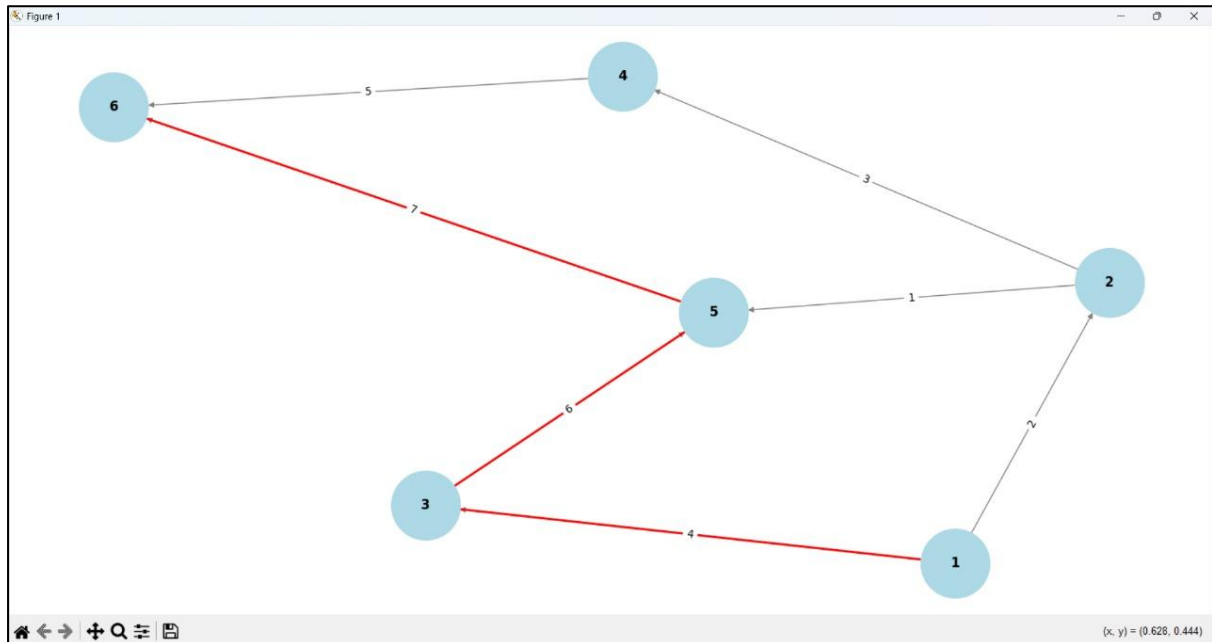
Enter times for activity 5-6:

Optimistic time (O): 3

Most likely time (M): 6

Pessimistic time (P): 15

Enter an activity (e.g., 1-2, 2-3) or type 'done' to finish: done



APPLICATIONS

The PERT-CPM Analysis Tool, which combines the Program Evaluation and Review Technique (PERT) and Critical Path Method (CPM), has a wide range of applications in project management across various industries. These methods are particularly valuable for projects that involve uncertainty, dependencies, and multiple tasks. Below are some of the key applications of the tool:

1. Construction Project Management

- **Project Scheduling:**

In construction projects, where multiple tasks must be completed sequentially and some tasks are dependent on others, the PERT-CPM Analysis Tool helps in accurately scheduling the tasks. By calculating the critical path, project managers can prioritize tasks and allocate resources efficiently.

- **Risk Mitigation:**

Construction projects often face delays due to unforeseen factors (e.g., weather, supply chain issues). Using PERT's probabilistic time estimation, project managers can plan for these uncertainties, create contingency plans, and reduce the risk of delays.

2. Software Development Projects

- **Task Dependency Management:**

In software development, tasks such as coding, testing, and debugging must follow a logical sequence. The tool helps manage these

dependencies, ensuring that each phase is completed on time and in the correct order.

- **Efficient Resource Allocation:**

The tool allows project managers to determine which tasks are critical and need immediate attention. It helps in optimizing resource allocation, ensuring that development teams focus on the most crucial aspects of the project, such as debugging and user testing.

3. Event Planning and Management

- **Timeline Optimization:**

Event planning involves coordinating multiple activities like venue booking, catering, decoration, and invitations. The tool helps break down these tasks, calculate the expected time for each, and identify the critical path, ensuring the event is organized efficiently and on time.

- **Budget and Resource Management:**

The CPM method allows event managers to allocate resources (e.g., budget, manpower) in a way that avoids delays in critical tasks. It also helps in identifying non-critical tasks that can be rescheduled to optimize resources.

4. Research and Development (R&D) Projects

- **Managing Complex Projects:**

R&D projects often involve a series of experiments, data collection, and analysis steps. The PERT-CPM Analysis Tool can be used to break down

the complex process into manageable tasks and estimate timeframes, helping R&D managers to keep the project on track.

- **Resource Planning:**

The tool helps allocate resources such as lab equipment, researchers, and materials to critical tasks. By identifying the critical path, the tool ensures that time-sensitive experiments are completed first, leading to timely results.

5. Manufacturing and Production Scheduling

- **Optimizing Production Line Efficiency:**

In manufacturing, multiple tasks such as assembly, testing, and packaging need to be done in a particular order. The PERT-CPM tool helps optimize production schedules by calculating the minimum time required to complete all tasks and identify bottlenecks.

- **Quality Control and Timely Delivery:**

By identifying critical tasks that directly impact the production timeline, the tool helps in ensuring quality control processes are conducted on time. Timely execution of tasks also results in meeting customer delivery deadlines.

CONCLUSION

The development and implementation of the PERT-CPM Analysis Tool has proven to be a significant advancement in the field of project management. By combining the strengths of both the Program Evaluation and Review Technique (PERT) and the Critical Path Method (CPM), this tool enables project managers

to efficiently plan, schedule, and manage complex projects across various industries. The ability to visualize task dependencies, estimate time frames, and optimize resources ensures that projects are executed on time and within budget, even in the face of uncertainties.

The tool's applications in sectors such as construction, software development, healthcare, event management, and manufacturing highlight its versatility and effectiveness in diverse environments. With its focus on identifying critical tasks, managing risks, and improving overall efficiency, the PERT-CPM Analysis Tool empowers project managers to make data-driven decisions and navigate the complexities of project execution.

In conclusion, the PERT-CPM Analysis Tool serves as an invaluable asset for modern project management, offering practical solutions to the challenges faced in coordinating and completing projects. By leveraging the power of this tool, organizations can achieve more predictable outcomes, enhance collaboration, and ensure the successful delivery of projects.

FUTURE SCOPE

The PERT-CPM Analysis Tool has many possibilities for improvement and wider use in the future. Here are some ideas for how it could evolve:

1. **Use of AI and Machine Learning:** By adding AI and machine learning, the tool could predict problems like delays and resource shortages based on

past projects. This would help managers make better decisions and avoid risks.

2. **Real-Time Updates:** The tool could connect to project management systems to update schedules automatically in real time. This way, the tool would always reflect the latest information about task progress and changes.
3. **Mobile App:** A mobile version of the tool could allow project managers to check and update the project schedule from their phones or tablets, making it easier to manage projects while on the go.
4. **Team Collaboration:** Adding features for team collaboration would allow multiple users to work on the same project, share updates, and communicate easily. This would improve coordination and teamwork.
5. **Integration with Other Tools:** The tool could work alongside other project management software, like Gantt charts or financial tracking systems, to create a more complete solution for managing all aspects of a project.
6. **Better Reports and Visuals:** Future versions could offer more advanced reporting features, such as customizable reports, interactive charts, and dashboards that make it easier to understand project data.
7. **Cloud-Based Version:** Moving the tool to the cloud would allow project teams to access and update the project information from anywhere, ensuring that everyone is on the same page.

With these improvements, the PERT-CPM Analysis Tool could become even more powerful, making project management easier, more efficient, and adaptable to different needs.

REFERENCES

- Python Documentation: <https://docs.python.org/>
- Scikit-learn Documentation: <https://scikit-learn.org/>
- TensorFlow Guide: <https://www.tensorflow.org/>
- Kaggle Datasets: <https://www.kaggle.com/datasets>
- Matplotlib Documentation: <https://matplotlib.org/stable/contents.html>

PRECEDENCE STUDY

1. Introduction

A **precedence study** is a crucial step in project scheduling that identifies the relationships between tasks or activities. By understanding the order in which activities must be performed, project managers can plan efficiently, allocate resources effectively, and anticipate potential delays. For this tool, which combines PERT and CPM, the precedence study helps define how activities in a project are interdependent and the critical paths that need to be managed.

2. Defining Activities and Their Dependencies

Each activity in the project must have defined relationships with other activities, specifying which activities must be completed before others can begin. These relationships are typically categorized into the following types:

- **Finish-to-Start (FS):** The most common type, where one activity must finish before the next can start.
- **Start-to-Start (SS):** The second activity can begin only after the first one starts.
- **Finish-to-Finish (FF):** The second activity can finish only after the first one finishes.
- **Start-to-Finish (SF):** The second activity can finish only after the first one starts (less common in practice).

The precedence study identifies and organizes these relationships, ensuring that the project schedule can be correctly constructed.

3. Identifying Critical Activities and Dependencies

In the context of PERT and CPM, it's essential to determine the **critical path** and the **float** (slack time) available for each activity. The critical path represents the longest path through the project, where any delay in the activities on this path will directly delay the project's overall duration.

- **Critical Path Method (CPM):** This focuses on identifying the longest sequence of activities that must be completed on time for the project to be finished on schedule.
- **Program Evaluation and Review Technique (PERT):** This method incorporates uncertainties in activity durations and estimates expected times using optimistic, pessimistic, and most likely estimates.

The precedence study must consider these techniques by identifying activities that have high uncertainty and those that are essential to the project's completion.

4. Flowchart of Dependencies

In this phase, the relationships between tasks can be represented as a flowchart or network diagram. A **precedence diagram** is typically used to map the activities and dependencies clearly. This network is then analyzed using the PERT or CPM methods to determine the critical path and overall project duration.

- **Nodes:** Represent the activities.
- **Arrows/Edges:** Represent dependencies between activities.

This study ensures that the tool can dynamically visualize the scheduling, taking into account the various dependencies between activities.

5. Tool Functionality

The dynamic scheduling tool must:

- Accept activity definitions (start-end tasks).
- Accept time estimates for each activity (optimistic, most likely, pessimistic).
- Generate a network diagram representing the dependencies and activity times.
- Calculate the **expected time** and **standard deviation** for each activity using PERT.
- Use **CPM** to identify the critical path.

- Dynamically update the schedule when activities or durations change, showing real-time effects on the critical path.

6. Conclusion

By performing the precedence study, we define the structure and logic that drives the scheduling tool. The relationships between activities, represented by the precedence diagram, guide the tool's ability to determine critical paths, update project timelines, and visualize the effects of any changes in real time. This step ensures that the tool can provide optimal scheduling solutions and assist project managers in making informed decisions about resource allocation, scheduling, and project control.

This precedence study provides a framework for understanding how the activities in our project relate to each other and how they should be scheduled. Integrating this with PERT and CPM in our tool will ensure a dynamic and accurate scheduling process.

LITERATURE SURVREY

Literature Survey on Dynamic Scheduling Using PERT and CPM

1. Introduction to Project Scheduling

Project scheduling is a key part of managing any project. It helps plan tasks, allocate resources, and ensure that the project is completed on time. Two common methods used for scheduling projects are **Program Evaluation and Review Technique (PERT)** and **Critical Path Method (CPM)**. While both methods

are used to organize projects, they have different approaches. PERT is often used for projects where task durations are uncertain, while CPM is used when task durations are known and predictable.

2. Program Evaluation and Review Technique (PERT)

PERT was developed in the 1950s to manage complex projects with uncertain timelines, like the U.S. Navy's Polaris missile program. PERT helps in estimating how long each task will take using three different time estimates:

- **Optimistic time (O):** The shortest time an activity could take.
- **Most likely time (M):** The most realistic estimate.
- **Pessimistic time (P):** The longest time an activity might take.

The expected time for an activity is calculated using:

$$E = \frac{O + 4M + P}{6}$$

This approach helps project managers account for uncertainty by providing a range of possible outcomes for each task.

Key References:

- **Malek, M., & Shoja, A. (2010):** "Project Scheduling with PERT and CPM" – discusses PERT's application in real-world projects.
- **Freeman, R. (1972):** "PERT: A Review of the Literature" – reviews the development and use of PERT.

3. Critical Path Method (CPM)

CPM, developed around the same time as PERT, is used when the time for each task is predictable. It focuses on finding the longest sequence of tasks that must

be completed on time to finish the project. This sequence is called the **critical path**. Any delay in tasks on this path will delay the entire project.

CPM also helps calculate **float time**, which shows how long a task can be delayed without affecting the overall project.

Key References:

- **Kerzner, H. (2017):** "Project Management: A Systems Approach to Planning, Scheduling, and Controlling" – discusses how CPM is used in managing projects.
- **Vasanth, S., & Ramesh, R. (2012):** "CPM and its Applications in Construction Projects" – explains how CPM is applied in construction projects.

4. Comparing PERT and CPM

Both PERT and CPM are useful for project scheduling, but they are suited to different types of projects. PERT is ideal for projects with uncertain task durations, such as research and development. CPM works best for projects with known task durations, like construction projects. Both methods have strengths and limitations, and choosing between them depends on the project's nature and the available data.

Key References:

- **Gupta, S., & Jain, A. (2006):** "A Comparative Study of PERT and CPM in Project Management" – compares the two methods and explains when to use each.

- **Morris, P. W. (2013):** "Managing Complex Projects: A New Model" – outlines the differences between PERT and CPM.

5. Dynamic Scheduling Using PERT and CPM

With the rise of project management software, PERT and CPM can now be used in more dynamic ways. These tools help adjust project schedules in real time based on changing conditions. For example, by combining PERT with risk analysis techniques, managers can predict delays and adjust the schedule accordingly. New technologies like **Artificial Intelligence (AI)** and **Machine Learning (ML)** are also being used to improve project scheduling by predicting potential issues and suggesting the best course of action.

Key References:

- **Cheng, M. Y., & Li, H. (2007):** "Dynamic Scheduling and Optimization of Complex Projects Using PERT and AI Techniques" – explores how AI is being used to improve project scheduling.
- **Smith, S., & Biddle, P. (2015):** "Using Machine Learning for Project Schedule Optimization" – discusses how machine learning can optimize project schedules.

CREATIVITY AND INNOVATION

In our project on developing a tool for dynamic scheduling using PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method), creativity and innovation play a key role in improving the efficiency and accuracy of project management processes.

Creativity in our project comes into play when designing the tool itself. Creating a user-friendly interface, coming up with new ways to visualize data, or simplifying complex scheduling processes through automation all require creative thinking. For instance, integrating different scheduling methods into one tool, allowing for dynamic changes as new data comes in, or developing a unique way to display project timelines could be seen as creative aspects of the project.

Innovation is demonstrated in how the tool transforms the traditional project management approach. While PERT and CPM are well-established techniques for scheduling, innovation in your project could mean creating a solution that allows for more dynamic, real-time adjustments. For example, implementing an automated way to update project schedules based on changing conditions or incorporating machine learning to predict delays and suggest adjustments could be considered innovative. The tool could innovate the way project managers interact with and make decisions using PERT and CPM, offering a more interactive, real-time system.

Why Creativity and Innovation Matter in our Project

1. **Enhancing Efficiency:** By creatively designing a tool that integrates PERT and CPM, you can offer a more streamlined and effective scheduling solution, reducing the time and effort required for manual scheduling.
2. **Adapting to Changing Environments:** In the dynamic nature of projects, the ability to adjust schedules on the fly is a key innovation. A tool that

can automatically accommodate changes and re-calculate schedules in real-time would be highly beneficial.

3. **User Experience:** Creative ideas in interface design and interaction ensure the tool is easy to use, helping project managers adopt the tool seamlessly without overwhelming them with complex calculations or data.
4. **Predictive Features:** Innovation could come in the form of predictive analytics, where the tool forecasts potential delays or issues and provides project managers with proactive solutions.

Conclusion

In the context of your project, creativity allows you to think outside the box when developing the tool, and innovation enables you to improve upon traditional scheduling methods. Together, they can help create a more powerful and adaptable project management tool, making scheduling more efficient, accurate, and dynamic in today's fast-paced project environments.

COMMUNITY IMPACT

The development of a dynamic scheduling tool using PERT (Program Evaluation and Review Technique) and CPM (Critical Path Method) has the potential to significantly impact both the professional community and society as a whole. Here's how it can make a difference:

1. **Improved Project Management Practices:** By providing an efficient tool for project scheduling, this tool can help project managers in various industries, such as construction, healthcare, and IT, to optimize project

timelines. With better scheduling, projects are more likely to be completed on time and within budget, benefiting businesses and clients.

2. **Increased Productivity and Efficiency:** The ability to dynamically update schedules based on changing conditions can improve productivity. Project teams can respond more quickly to delays or unforeseen circumstances, reducing wasted time and resources. This can have a ripple effect, enhancing the productivity of entire industries.
3. **Accessible for Small and Medium Enterprises (SMEs):** Many small and medium-sized businesses (SMEs) struggle with advanced project management due to limited resources. A tool that simplifies and automates scheduling using PERT and CPM can level the playing field, allowing SMEs to take advantage of sophisticated project management techniques that were previously available only to larger organizations.
4. **Educational Impact:** The tool can be used in educational settings to teach students about PERT, CPM, and project management principles. By making complex scheduling concepts more accessible, students and professionals can gain hands-on experience with practical tools, bridging the gap between theory and real-world application.
5. **Supporting Sustainable Development:** In industries like construction, energy, and healthcare, improving project timelines and resource allocation can lead to more sustainable practices. Projects completed on time with fewer delays reduce environmental impact, minimize resource waste, and contribute to more sustainable practices in the long run.

6. **Enhanced Decision-Making:** The dynamic nature of the tool empowers project managers and stakeholders to make better-informed decisions. They can easily see the consequences of schedule changes, and adjust resources or timelines accordingly, leading to better decision-making and smoother project execution.
7. **Job Creation and Economic Growth:** As businesses and industries adopt more efficient project management practices, there is potential for job creation in fields like software development, data analysis, and project management. Moreover, the economic growth of industries that adopt such technologies will benefit local economies.

Conclusion

The community impact of a dynamic scheduling tool goes beyond individual projects; it has the potential to improve project management practices across industries, increase productivity, and support the growth of small businesses. Through education and sustainable development, the tool can have a lasting positive influence on both professionals and society, driving growth, efficiency, and better decision-making.